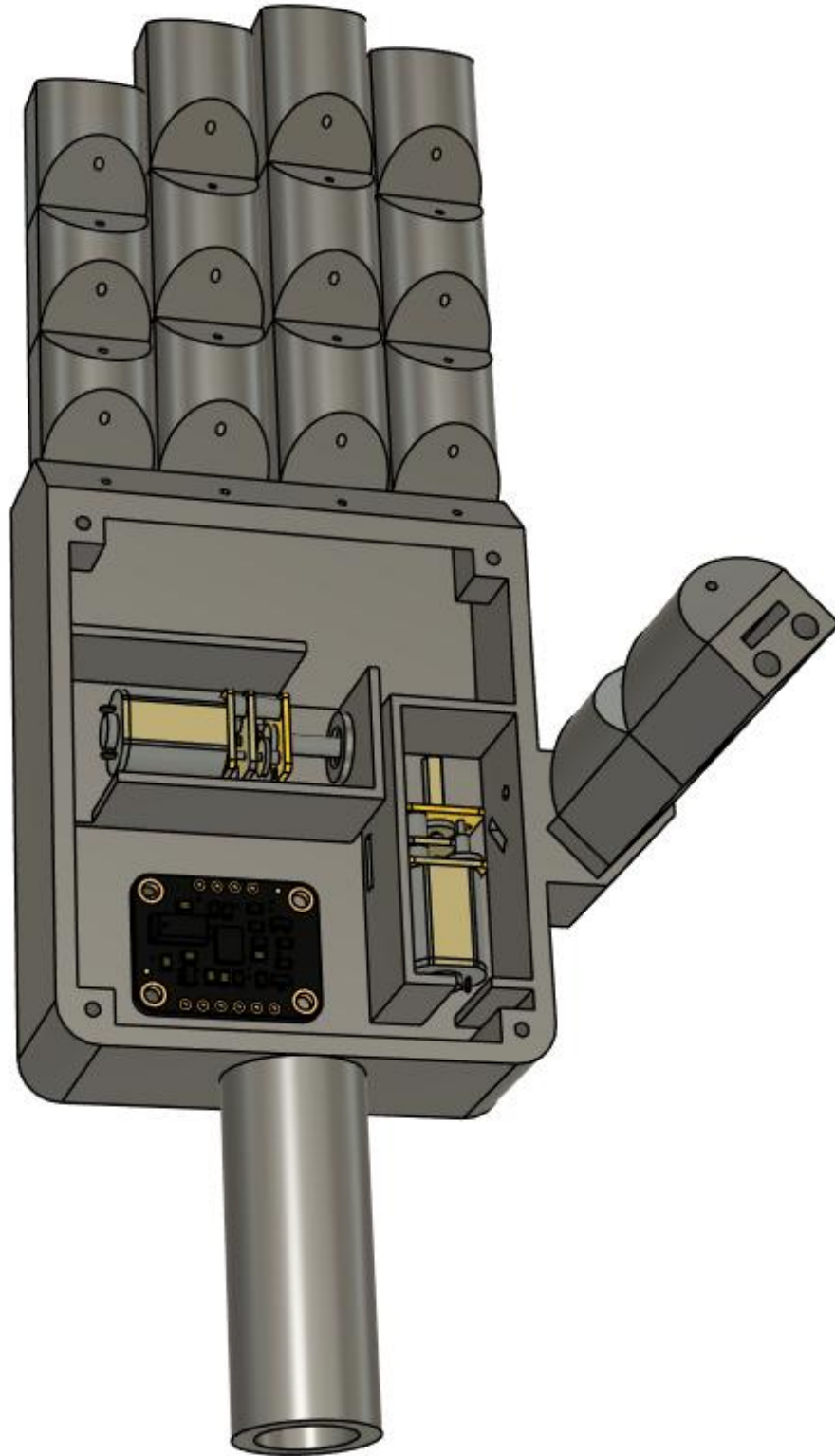


CAMERON FRACH

ENGINEERING PORTFOLIO

Biomedical Engineering | University of Surrey

CAD • FEA • Embedded Control • Prototyping



Selected academic engineering work | 2024 to 2026

Engineering Profile

SKILLS • PROJECTS • METHODS

Biomedical Engineering student with practical experience in CAD, embedded control, FEA, MATLAB, manufacture and testing. The projects here show the work I produced and the changes I made when testing exposed a problem.

CAD AND PRODUCT DESIGN

Fusion 360 assembly modelling
Engineering drawings
Dimensional refinement
FDM design for manufacture

ANALYSIS

ANSYS Mechanical and Workbench
Fusion 360 static stress
MATLAB numerical modelling
Engineering calculations

EMBEDDED SYSTEMS

Arduino and C++
ADC sensor processing
PWM motor control
L293D H bridge actuation

TESTING AND PROTOTYPING

FDM 3D printing
Component integration
Black box and white box testing
Experimental measurement

SELECTED WORK

01

Tendon Driven Prosthetic Hand

CAD • FEA • 3D PRINTING • TESTING

Designed and refined printed components for a child sized prosthetic hand and contributed to system integration and testing.

02

Embedded Motor Control System

ARDUINO • C++ • SENSORS • CONTROL

Built and programmed a light controlled DC motor system, then diagnosed sensor jitter and other faults through structured testing.

03

Composite Structural Analysis

ANSYS • CFRP • OPTIMISATION

Reduced the mass of a CFRP sandwich structure while checking deformation, stress, factor of safety and mesh convergence.

04

Mechanical CAD and Drawing

FUSION 360 • DFM • DRAWINGS

Modelled a mechanical hoist assembly and produced the corresponding engineering drawing with dimensions, views and item callouts.

05

Engineering Analysis

MATLAB • MOTION CAPTURE • STRAIN GAUGES

Applied numerical and experimental methods to gait analysis, motion capture and beam bending.

The following pages focus on the engineering evidence: CAD, code, analysis outputs, test results and the design changes that followed.

01 Tendon Driven Prosthetic Hand

Mechanical design, manufacture, FEA, integration and testing

02 Embedded Motor Control System

Arduino, C++, sensor processing, control logic and debugging

03 Composite Structural Analysis and Optimisation

ANSYS, CFRP sandwich structure, mesh convergence and mass reduction

04 Mechanical CAD and Engineering Drawings

Fusion 360 assembly modelling and technical communication

05 Engineering Analysis and Experimentation

Gait modelling, motion capture and beam stress measurement

Tendon Driven Prosthetic Hand

DESIGN • MANUFACTURE • TEST

A three month team project to develop a low cost articulated prosthetic hand for a 12 year old user. The final system combined a 3D printed mechanical structure, tendon actuation, two gear motors, force sensing, acceleration sensing and Arduino control.

180 g

final hand mass

~19.6 N

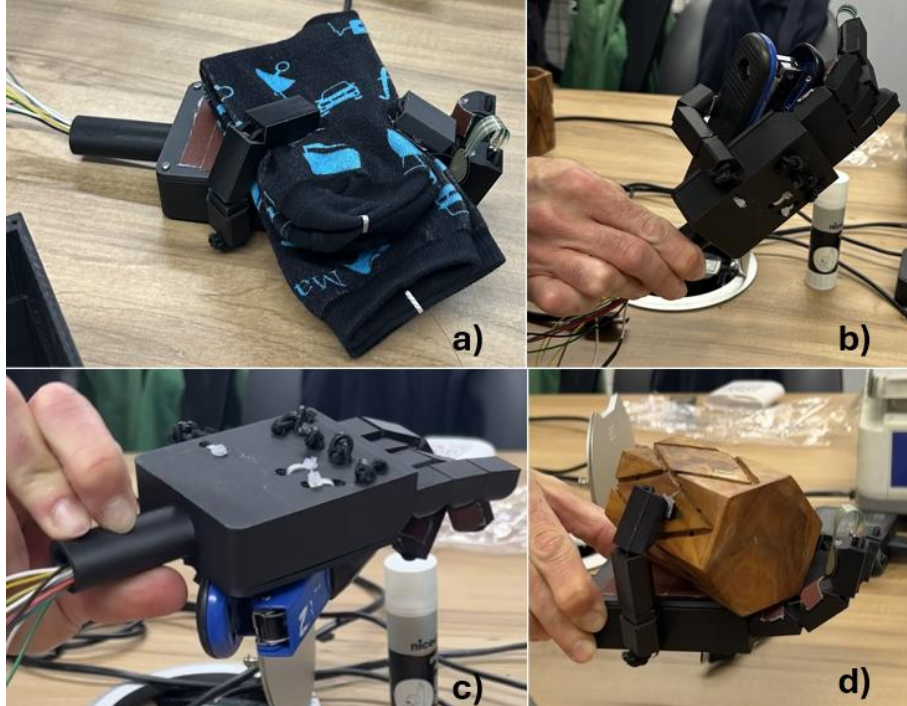
grip force reference

1.5 m

drop test survived

£211.34

total project cost



FINAL PROTOTYPE DURING FUNCTIONAL TESTING

MY CONTRIBUTION

I developed and refined the CAD geometry in Fusion 360, produced 3D printed hand and control components from prototype through final design, improved dimensions for manufacture, and contributed to planning and system integration.

FINAL SYSTEM

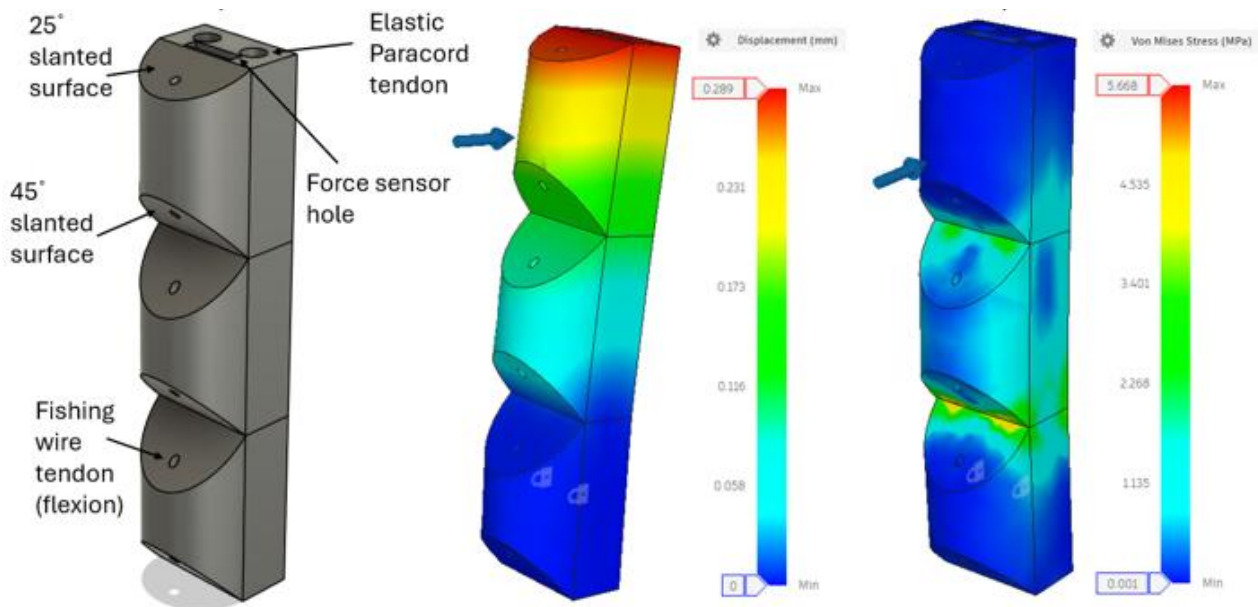
Fishing wire tendons transmitted motor force to the fingers while elastic paracord provided passive return. Force and acceleration sensing supported safe operation and an automated shutdown function.

VALIDATION

The finished hand gripped several everyday objects and still worked after the 1.5 m drop test. These were two of the main checks we used to see whether the prototype worked as intended.

Prosthetic Hand: Development and Analysis

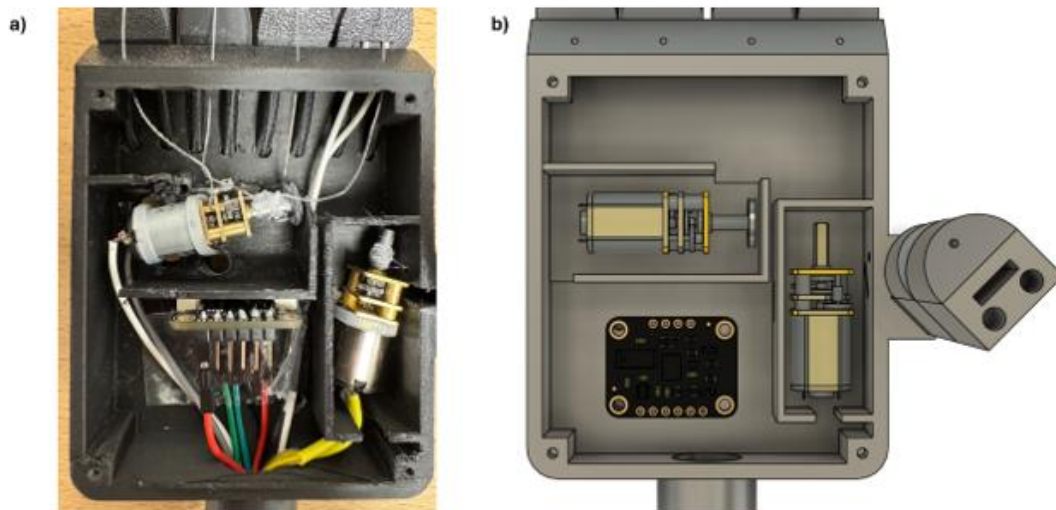
ENGINEERING DEVELOPMENT



FINGER DESIGN AND STRUCTURAL ASSESSMENT

STRUCTURAL ASSESSMENT

A Fusion 360 static stress study applied a 20 N load at the fingertip to check whether the finger geometry could support the target grip force. Maximum displacement was approximately 0.289 mm. The stress field showed the highest local concentrations around the joint transitions, which were therefore important areas when reviewing the geometry.



PHYSICAL INTERNAL ASSEMBLY AND CAD PACKAGING

DESIGN CHANGES DRIVEN BY MANUFACTURE AND TESTING

1. The thumb base angle changed from 90° to 45° after testing showed that improved opposition was needed for more stable grasping.
2. Tendon routing holes were enlarged after FDM printing produced tighter clearances than the CAD model suggested.
3. A pulley concept was replaced by direct motor shaft routing after the tendon repeatedly tangled during operation.
4. Palm geometry was revised through several prototypes to package the motors, sensor hardware and wiring while retaining usable printed parts.

Embedded Motor Control System

ARDUINO • SENSORS • CONTROL

I designed and programmed a manual and light controlled DC motor system using an Arduino Uno, LCD keypad shield, phototransistor and L293D motor driver. The project required sensor calibration, user input handling, direction control, PWM speed control and a clear operating interface.

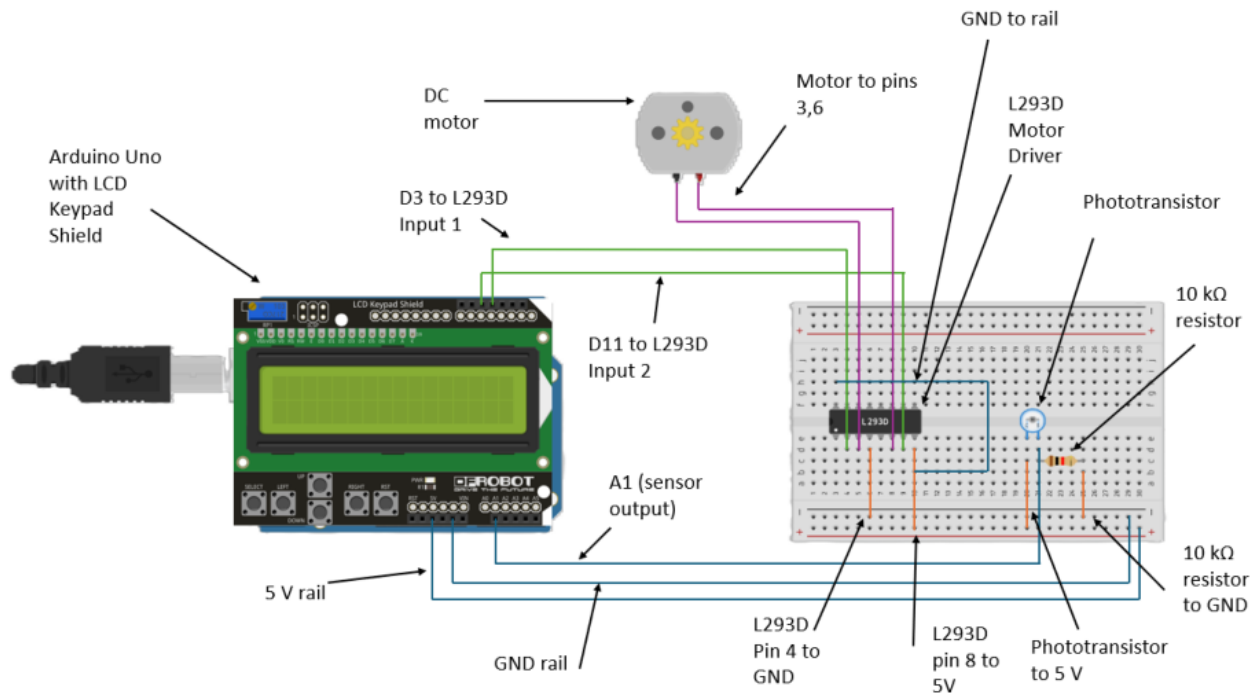


Figure 1: Circuit diagram of the light-controlled DC motor system using Arduino Uno, LCD keypad shield, phototransistor sensor and L293D motor driver

SYSTEM CIRCUIT AND HARDWARE INTERFACES

CONTROL ARCHITECTURE

The phototransistor was read through a 10 bit ADC input. In manual mode, keypad commands changed speed and direction directly . In automatic mode, the sensor reading was compared with a calibrated ambient reference. Readings above or below the reference commanded opposite motor directions, while the distance from the reference was mapped to the 0 to 255 PWM range.

10 bit ADC resolution	2 modes manual and automatic	PWM speed command	L293D bidirectional drive
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OUTPUT STAGE

The software converted the selected speed and direction into two PWM outputs for the H bridge. One input received the PWM command while the opposite input was held low. When the required speed was zero, both outputs were set to zero so the motor remained stationary.

Embedded Control: Code and Verification

C++ • DEBUGGING • TESTING

AUTOMATIC CONTROL LOGIC

The central control problem was preventing small sensor fluctuations around the ambient reference from causing rapid changes in motor direction. I introduced a tolerance band around the calibrated ambient value and refined it through repeated testing until the response was stable.

```
int tolerance = 70;

if (rawLight < ambientRaw - tolerance) {
    direction = -1;
    motorSpeed = map(rawLight, ambientRaw - tolerance,
                     sensorMin, 0, 255);
    motorSpeed = constrain(motorSpeed, 0, 255);
}
else if (rawLight > ambientRaw + tolerance) {
    direction = 1;
    motorSpeed = map(rawLight, ambientRaw + tolerance,
                     sensorMax, 0, 255);
    motorSpeed = constrain(motorSpeed, 0, 255);
}
else {
    motorSpeed = 0;
}
```

TEST, DIAGNOSE, REFINE

Motor jitter: Small changes in the light sensor repeatedly crossed the ambient reference. I traced the behaviour to sensor noise and introduced the tolerance band, increasing it during testing until a value of ± 70 ADC counts produced stable operation.

Display scaling: Mapped light values could exceed 100 percent when the sensor moved outside the calibrated range. I constrained the displayed value to the valid 0 to 100 percent interval.

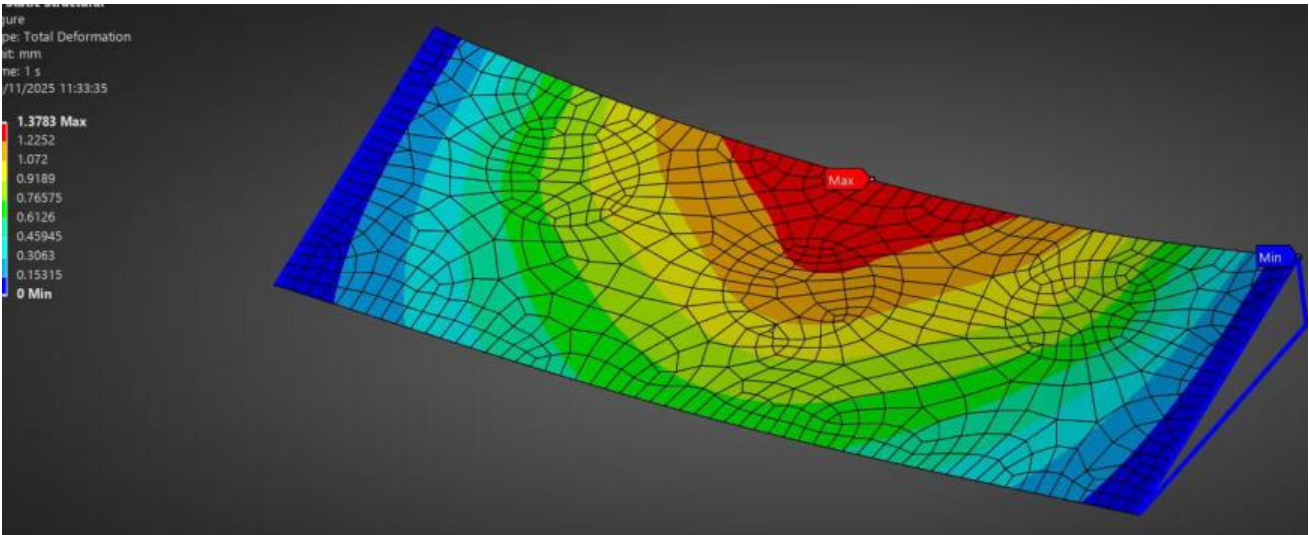
LCD output: Shorter status messages left characters from the previous display state. I cleared the residual characters by formatting the replacement text consistently.

Verification: Black box testing checked mode switching, manual control, automatic light response and LCD behaviour. White box testing then checked the internal decision paths for mode selection, keypad input, light control, speed and direction.

Composite Structural Analysis and Optimisation

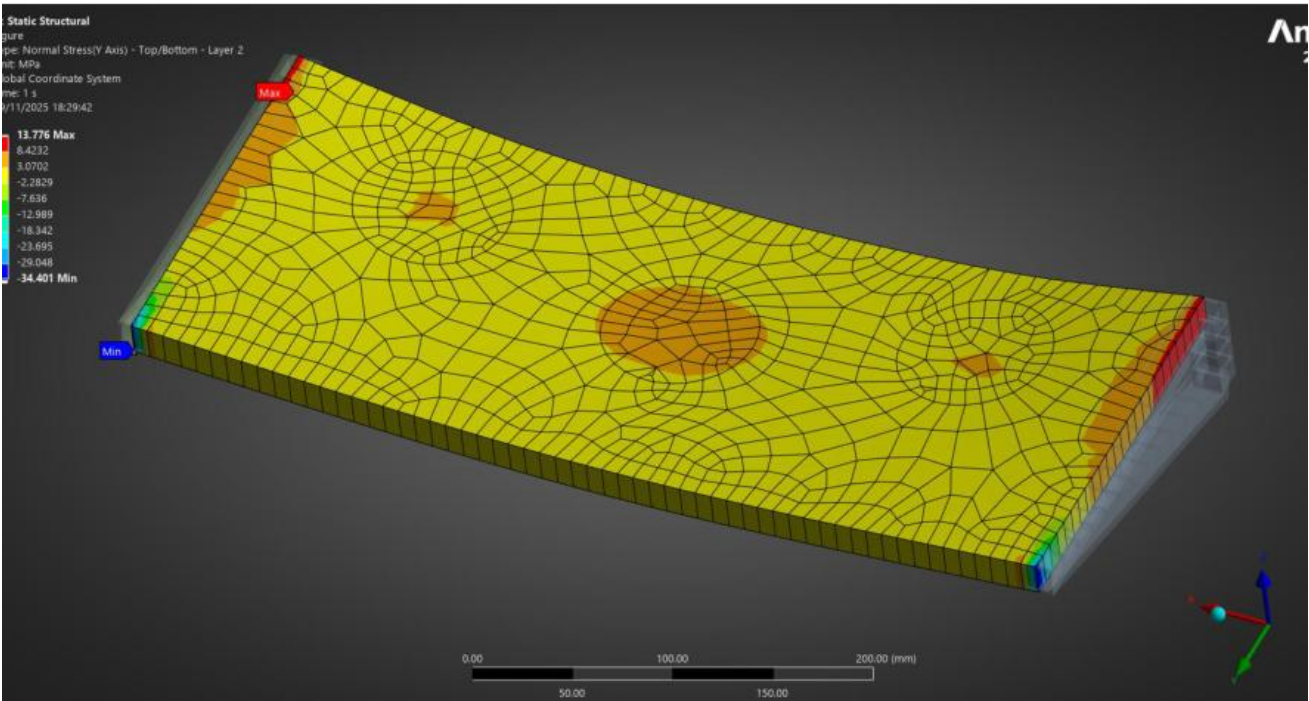
ANSYS MECHANICAL

I modelled a CFRP sandwich shelf and steel support structure under a 3750 N design load. The objective was to reduce mass while keeping maximum deflection below 5 mm and maintaining factors of safety above 1.



FINAL MODEL TOTAL DEFORMATION

The improved model produced a maximum deformation of 1.38 mm. This remained well below the 5 mm design limit, leaving margin for the mass reduction while preserving the required stiffness.



FINAL MODEL STRESS DISTRIBUTION

I evaluated deformation and ply stresses for the CFRP skins and foam core, compared the initial and improved geometries, and calculated factors of safety. The global mesh size was refined from 10 mm to 5 mm. The resulting change in maximum deformation was below 2 percent, providing a convergence check on the result.

1.82 to 1.59 kg total mass	~13% mass reduction	1.38 mm maximum deflection	<2% mesh result change
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ENGINEERING DECISION

The final geometry reduced mass by about 13 percent while staying below the 5 mm deflection limit. Refining the mesh from 10 mm to 5 mm changed maximum deformation by less than 2 percent.

Mechanical CAD and Engineering Drawings

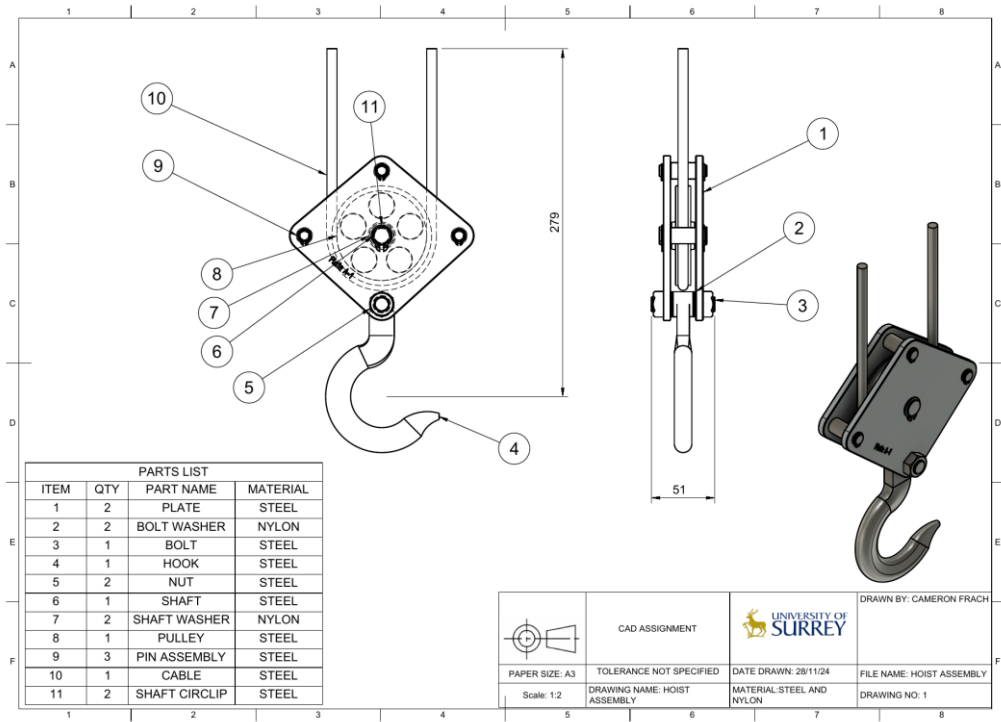
FUSION 360 • DESIGN FOR MANUFACTURE

I developed a multi component mechanical hoist assembly in Fusion 360 and produced the associated engineering drawing. The work focused on modelling component relationships accurately and communicating the assembly in a form that could support manufacture and inspection.



3D ASSEMBLY MODEL

The assembly model was used to check the overall geometry, orientation of the hook and supporting plates, and the relationship between pins, fasteners and the surrounding structure.



ASSEMBLY ENGINEERING DRAWING

I then produced the engineering drawing from the same assembly, using orthographic and isometric views, dimensions, item callouts and a parts list so the component relationships could be inspected clearly.

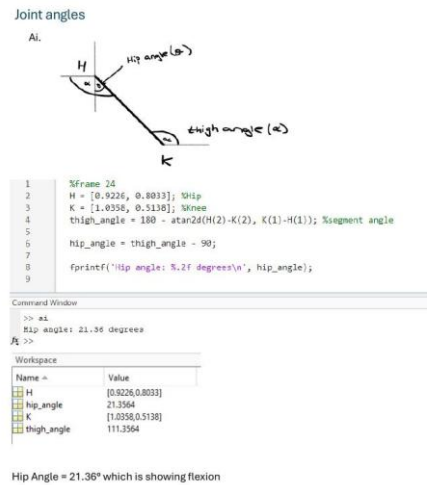
Engineering Analysis and Experimentation

MATLAB • TESTING • BIOMECHANICS

Computational Gait Analysis MATLAB • KINEMATICS

I used marker coordinates to calculate lower limb segment angles in MATLAB. This example shows my hip calculation for one frame, including the geometry, code and 21.36° flexion result. I then repeated the method for the knee and ankle and extended the coursework to joint moments and power.

Biomechanics – 2D analysis of gait

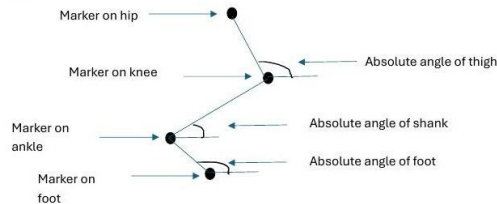


Coursework extract showing my geometry, MATLAB calculation and output.

Motion Capture Biomechanics QUALISYS • STATISTICS

I compared rearfoot and forefoot running using an 11 camera Qualisys system at 250 Hz. Markers at the hip, knee, ankle and foot were used to calculate segment and joint angles. Paired t tests found significant differences in absolute foot angle and relative ankle angle between two segments.

Figure 1. Markers placed on participants showing absolute angles



Marker positions and absolute segment angles used in the analysis.

Beam Stress Experiment STRAIN GAUGES • BENDING

I measured strain across a T section beam during four point bending and converted it to stress using Hooke's law. The graph shows the measured stress distribution for four applied loads. The experimental neutral axis was about 16 mm, compared with a theoretical value of 18 mm.

